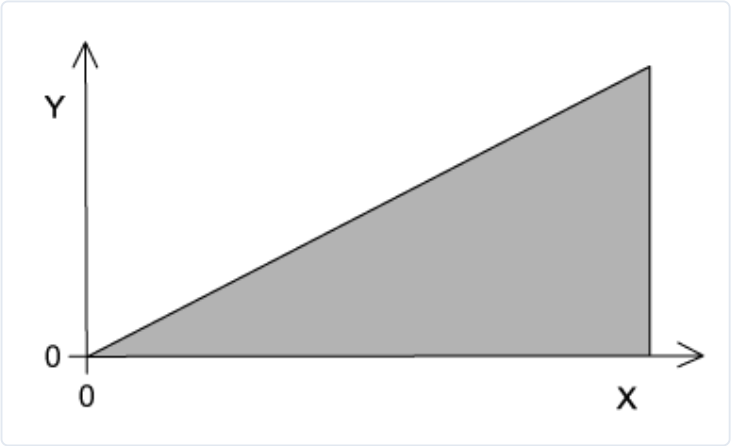
CIE AS PHYSICS (9702) • QUESTION PAPER

# 6.2 Elastic & Plastic Behaviour

**Syllabus Section:** 6. Deformation of Solids **Paper Type:** Multiple Choice (Paper 1) **Total:** 25 Questions (25 Marks)

## Question 1 Easy [1 mark]

A student experimented on a metal wire; the graph below was plotted.




The area under the graph shows the total strain energy stored in the wire when stretched.

How should the axes be labelled?

| | Y | X |
|---|---|---|
| **A** | mass | extension |
| **B** | force | extension |
| **C** | strain | energy |
| **D** | stress | strain |

**A**

**B**

**C**

**D**

**Question 2** Easy [1 mark]

Four rods were made from four different materials; all the rods had the same dimensions.

At room temperature, which of the following would sustain the biggest plastic deformation?

**A** the ductile material steel

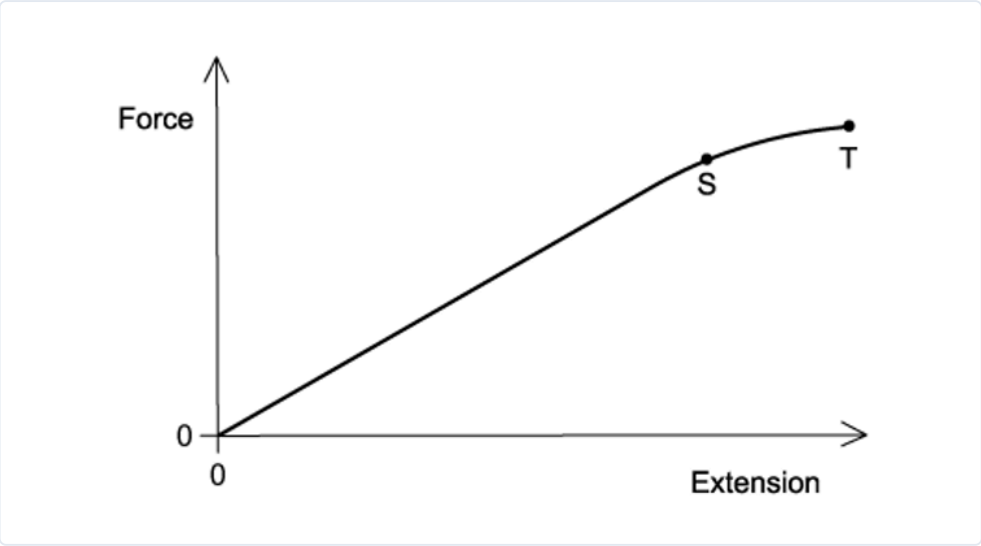
**B** the brittle material glass

**C** the brittle material carbon

**D** the ductile material aluminium

**Question 3** Easy [1 mark]

A student drew a force-extension graph for a sample of metal. She subjected the sample to a force that increased to a maximum then decreased to zero.




When the sample contracts it follows the same force-extension curve as when it was being stretched.

What is the behaviour of the metal between S and T?

**A** both elastic and plastic

**B** elastic but not plastic

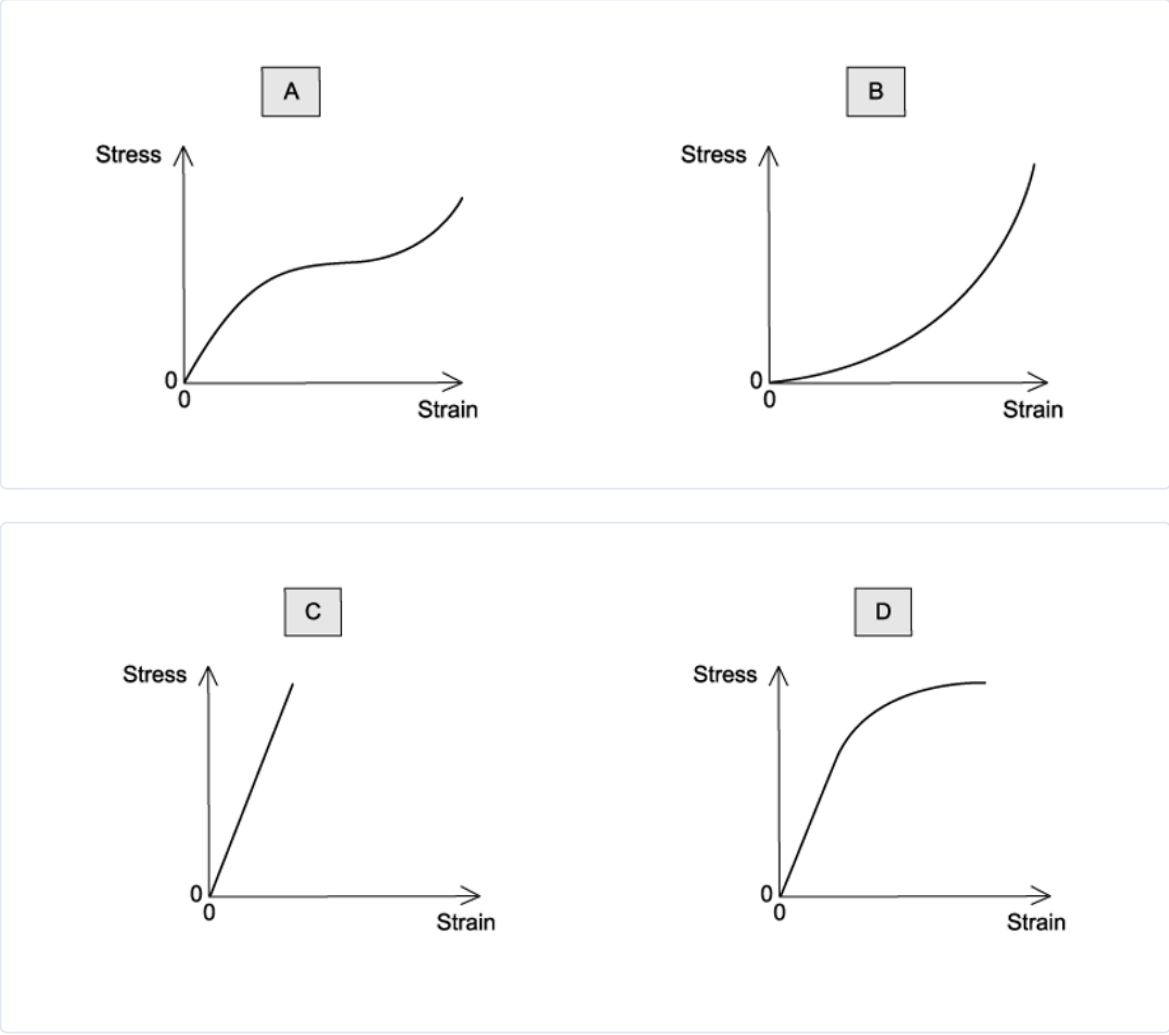
**C** not elastic and not plastic

**D** plastic but not elastic

**Question 4** Easy [1 mark]

Graphs of the stress-strain for four different materials are shown below.

Which one shows the correct graph for a ductile metal ?




A

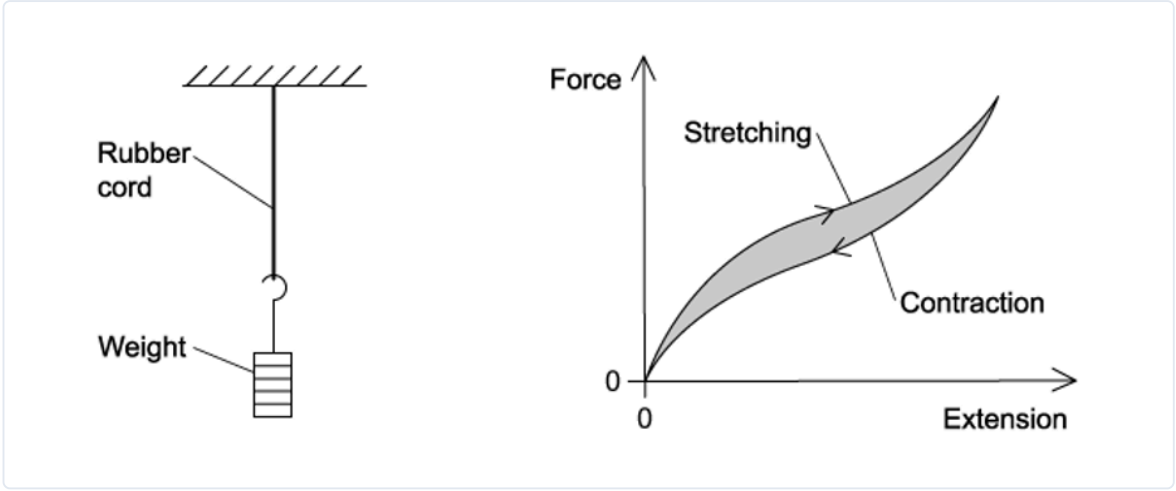
B

C

D

**Question 5** Easy [1 mark]

A weight was attached to the lower end of a length of rubber. The rubber was hung from a solid support. The weight was gradually increased from zero then gradually reduced to zero.




The graph shows the force-extension curve for contraction (below) and the force-extension curve for stretching.

What does the shaded area between the curves represent?

**A** the amount of elastic energy stored in the rubber

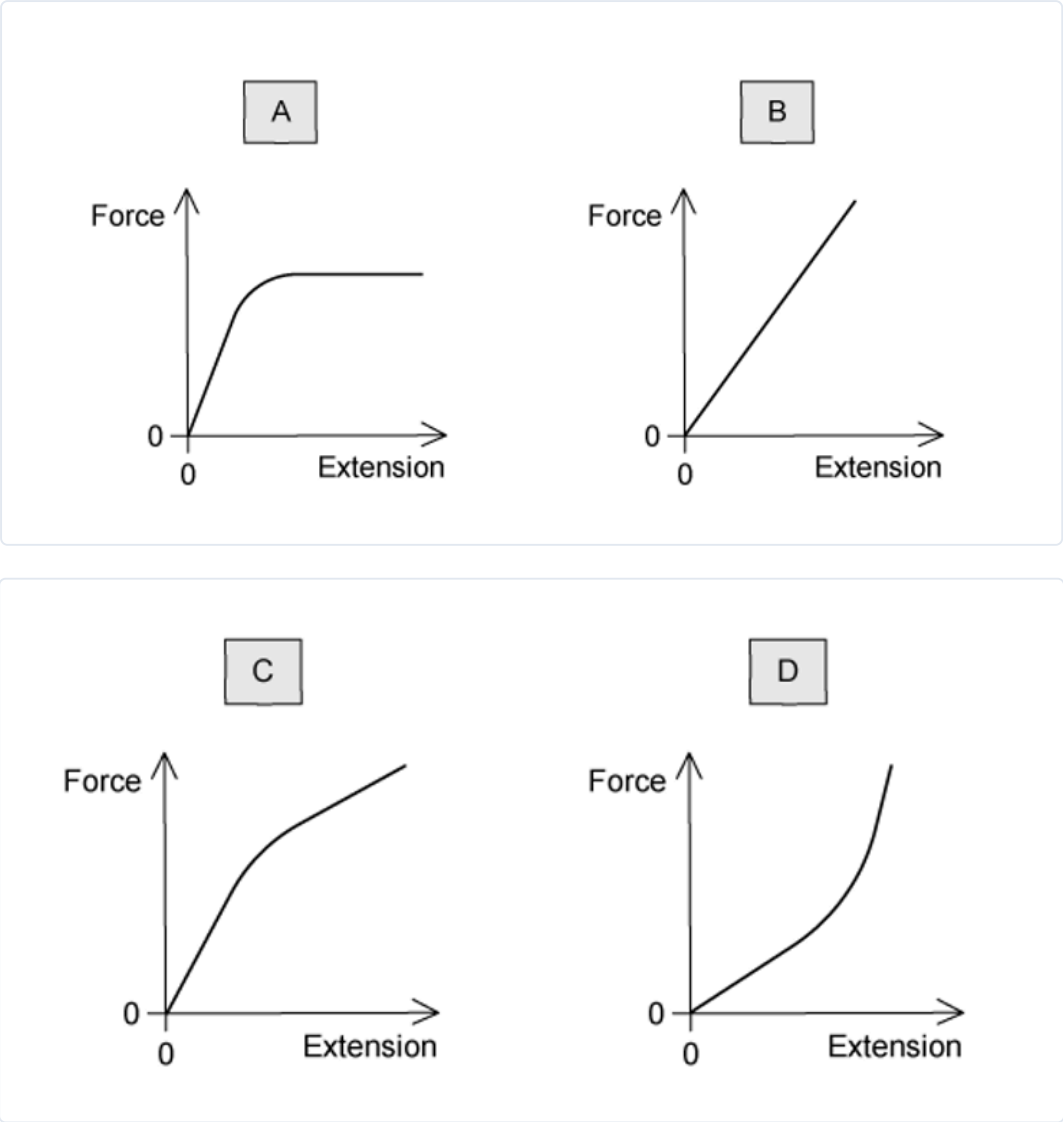
**B** the work done by the rubber cord during contraction

**C** the work done on the rubber cord during stretching

**D** the amount of thermal energy dissipated in the rubber

**Question 6** Easy [1 mark]

The graphs below are force-extension graphs, which one is a rubber band this is stretched almost to its breaking point ?




A

B

C

D

## Question 7 Easy [1 mark]

A metal wire had a tension *F* and extension *x* with a Young modulus of *E*; the wire obeyed Hooke's law

Which expression represents the elastic strain energy stored in the wire?

**A** $\frac{1}{2}Fx$

**B** $Ex$

**C** $\frac{1}{2}Ex$

**D** $Fx$

## Question 8 Easy [1 mark]

Which of the following is the correct definition for elastic and plastic behaviour?

| | Elastic behaviour of a material | Plastic behaviour of a material |
|---|---|---|
| **A** | returns to its original shape and size | suffers permanent deformation |
| **B** | obeys Hooke's Law | extends continuously under a steady load |
| **C** | has a linear force-extension curve | has a horizontal force-extension curve |
| **D** | extends only within the limit of proportionality | extends beyond the limit of proportionality |

**A**

**B**

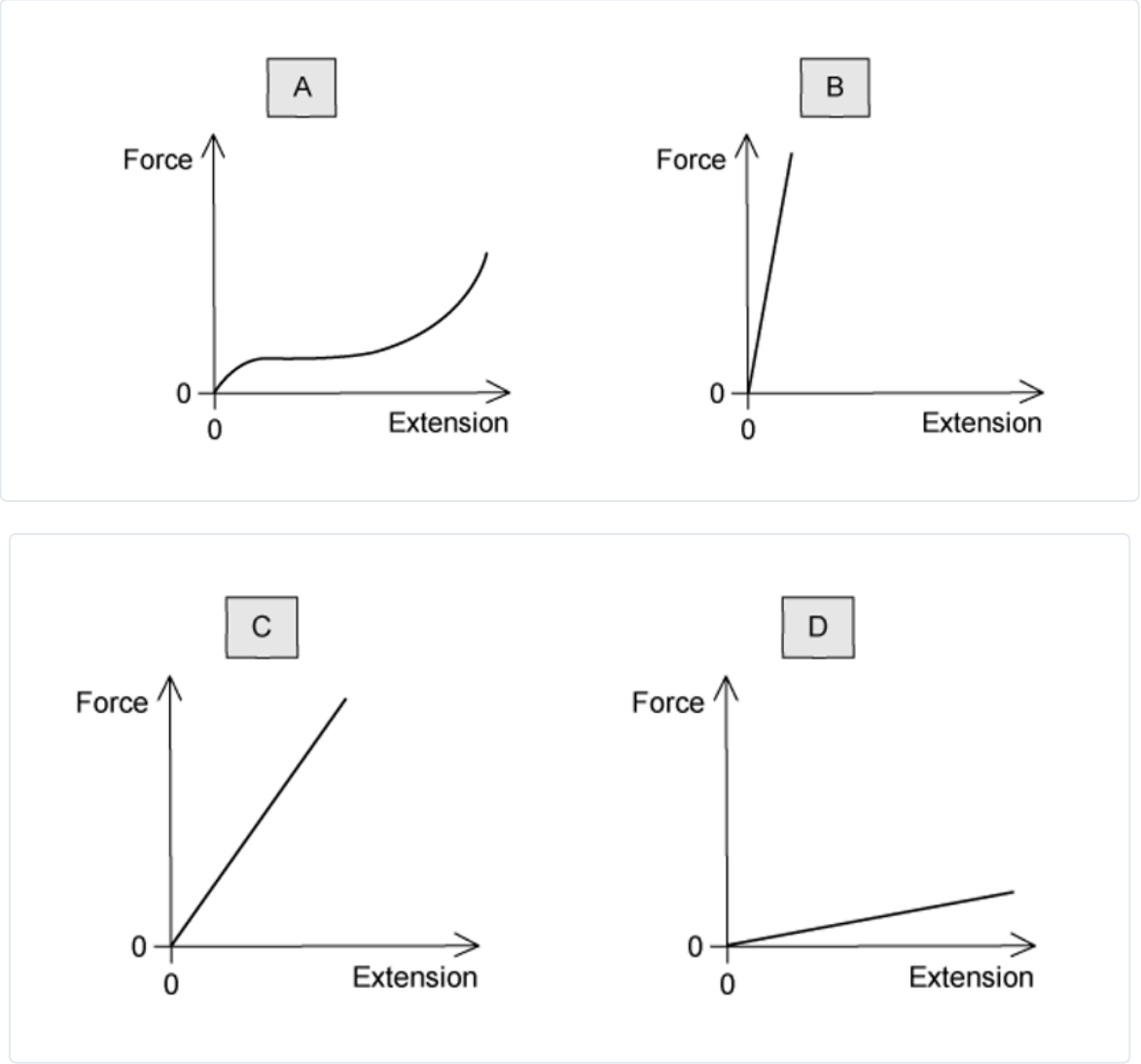
**C**

**D**

**Question 9** Easy [1 mark]

The graphs below all show force-extension graphs; they are all drawn to the same scale.

Which graph shows the object that has been deformed and has the greatest amount of elastic potential energy ?




A

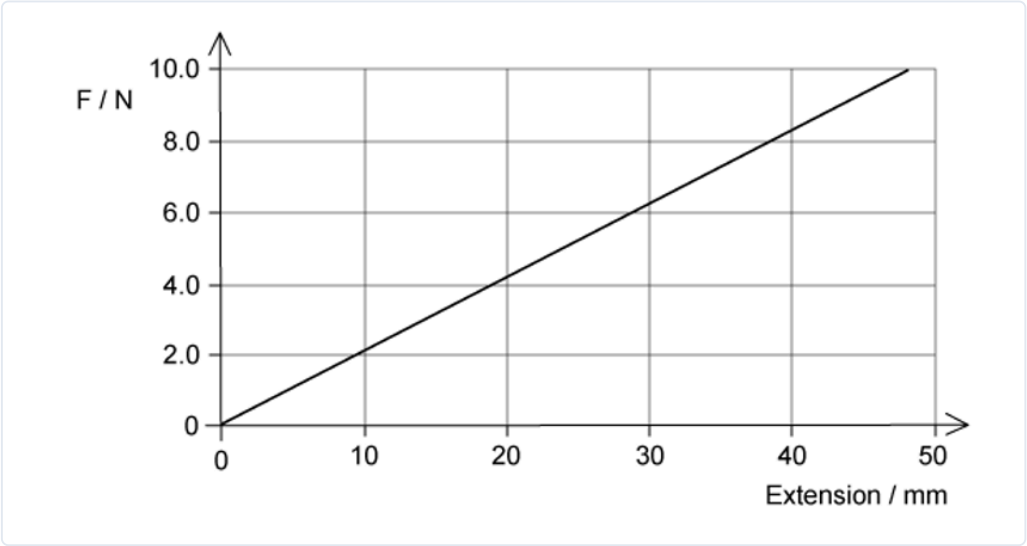
B

C

D

**Question 10** Easy [1 mark]

A spring is extended with varying forces; the graph below shows the results.




What is the energy stored in the spring when the extension is 30 mm?

**A** 0.25 J

**B** 0.95 J

**C** 0.095 J

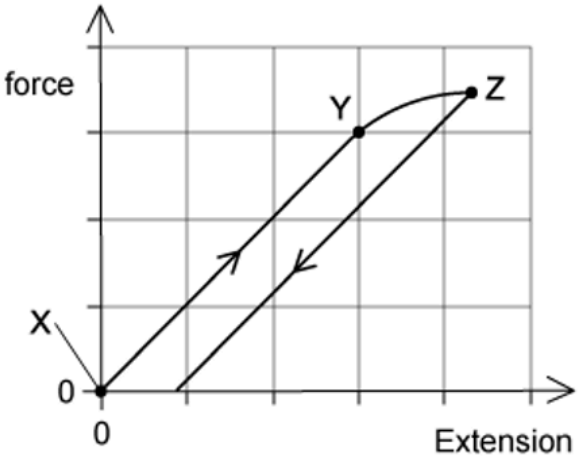
**D** 0.19 J

**Question 11** Medium [1 mark]

A student was investigating the extension in a long thin metal wire. They suspended the wire from a fixed support, and hung it vertically then hung masses from the lower end.

The load was increased and decreased from zero to the maximum and back again.

The graph shows the force-extension of the wire.




Where on the graph would the elastic limit be found?

**A** exactly at point Z

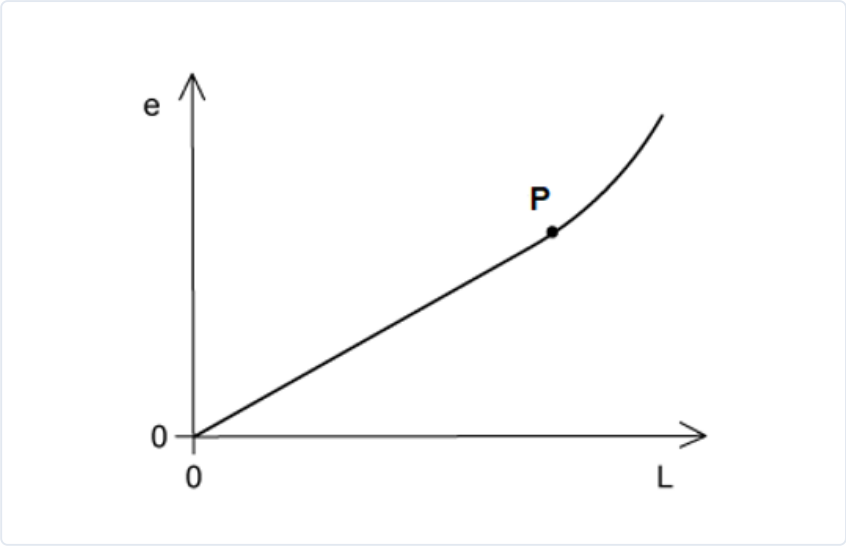
**B** exactly at point Y

**C** just beyond point Y

**D** anywhere between point X and point Y

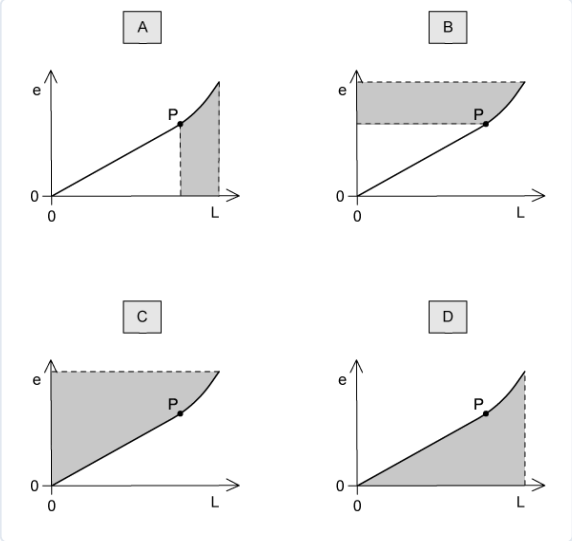
## Question 12 Medium [1 mark]

A rod is extended with a force so that its length increases. The variation with load *L* of the extension *e* of the rod is shown in the graph.




Point **P** is the elastic limit.

Which shaded area represents the work done during the plastic deformation of the rod ?




**A**

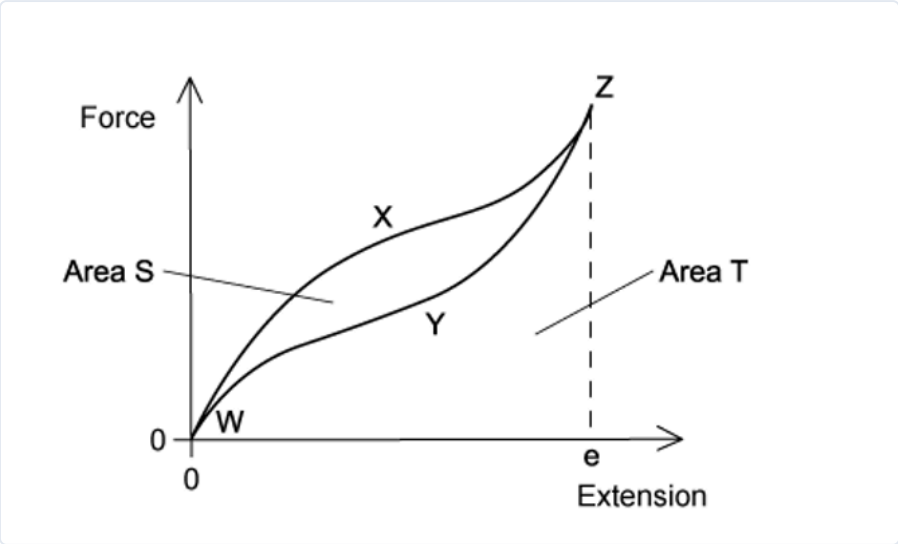
**B**

**C**

**D**

**Question 13** Medium [1 mark]

A student was investigating a rubber band when stretched and relaxed to its original length. The graph shows the force-extension of the rubber band.




As the force is increased, the curve follows the path WXZ to extension *e*. As the force is reduced, the curve follows the path ZYW to return to zero extension.

The area labelled S is between the curves WXZ and ZYW. The area labelled T is bounded by the curve ZYW and the horizontal axis.

Which statement about the process is correct?

**A** area S is the energy which heats the band as it is stretched to *e*

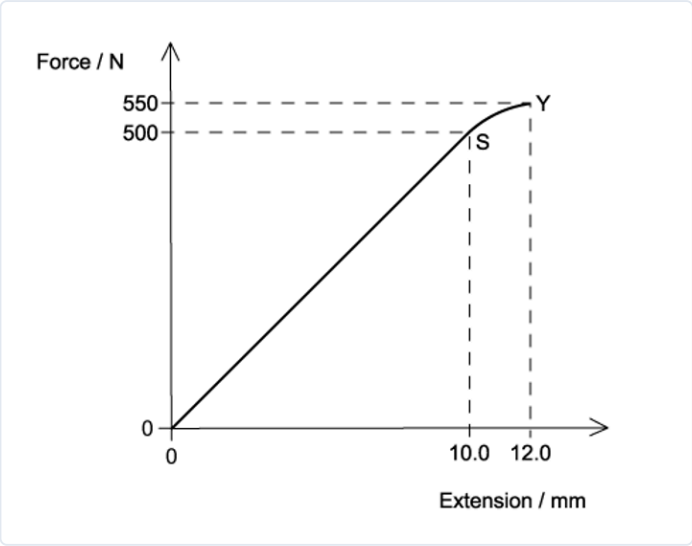
**B** area S is the elastic potential energy stored in the band when it is stretched to *e*

**C** (area S + area T) is the minimum energy required to stretch the band to *e*

**D** area S is the spring constant of the band at e

**Question 14** Medium [1 mark]

A sample of metal was stretched until it started to go into plastic deformation. The graph below shows the behaviour of the metal.




When the sample was stretched from 0 to 12.0 mm extension, what is the work done? Assume that the curve SY is a straight line to simplify the calculation

**A** 6.60 J

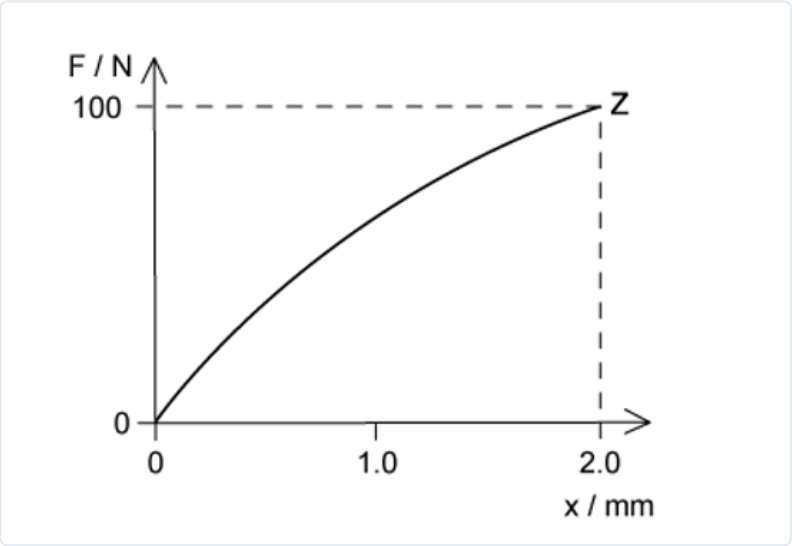
**B** 3.60 J

**C** 3.55 J

**D** 3.30 J

**Question 15** Medium [1 mark]

A wire made from a new composite material was loaded with masses and the force-extension curve drawn.




What could be the value of the strain energy stored in the wire when it is stretched elastically to point Z ?

**A** 0.20 J

**B** 0.11 J

**C** 0.10 J

**D** 0.09 J

## Question 16 Medium [1 mark]

A student was investigating a spring with a spring constant $k$ of 3.0 N $cm^{-1}$, the spring was stretched over a range in which elastic deformation occurred.

Which row, for the stated applied force, gives the correct extension and strain energy?

| | Force / N | Extension / cm | Strain energy / mJ |
|---|---|---|---|
| **A** | 24.0 | 8.0 | 960 |
| **B** | 12.0 | 3.0 | 200 |
| **C** | 6.0 | 2.0 | 120 |
| **D** | 3.0 | 1.0 | 1.5 |

**A**

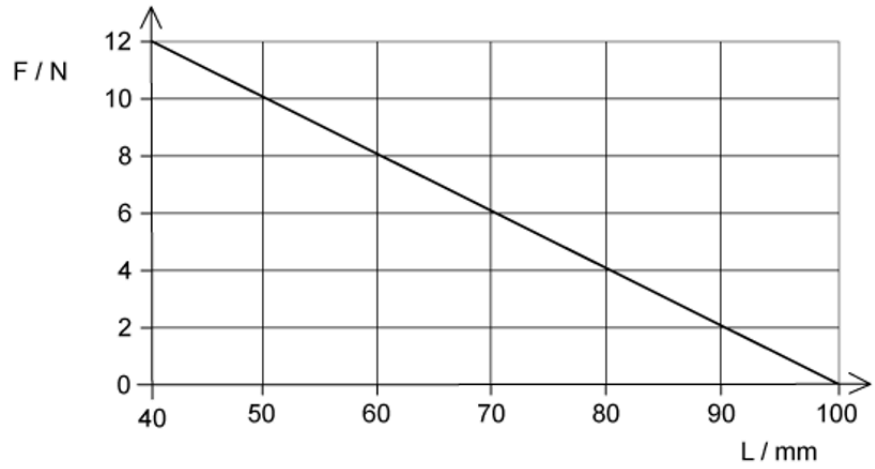
**B**

**C**

**D**

**Question 17** Medium [1 mark]

A force compresses a mattress spring by 100 mm. The graph shows the effect of the force *F* on the length *l* of the spring.




What is the energy stored in the spring when the length is 70 mm?

**A** 0.63 J

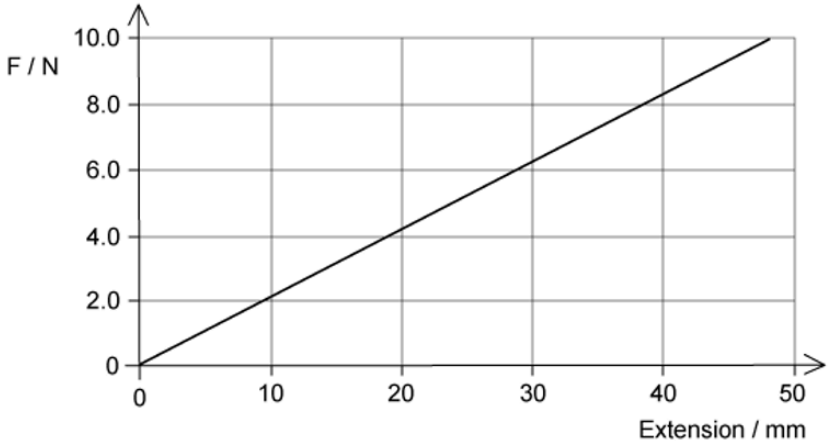
**B** 0.27 J

**C** 0.21 J

**D** 0.090 J

**Question 18** Medium [1 mark]

A spring is extended due to a force. The graph shows the effect of changing force on the extension of the spring.




What is the elastic potential energy of the spring with a force of 8.0 N ?

**A** 152 mJ

**B** 58 J

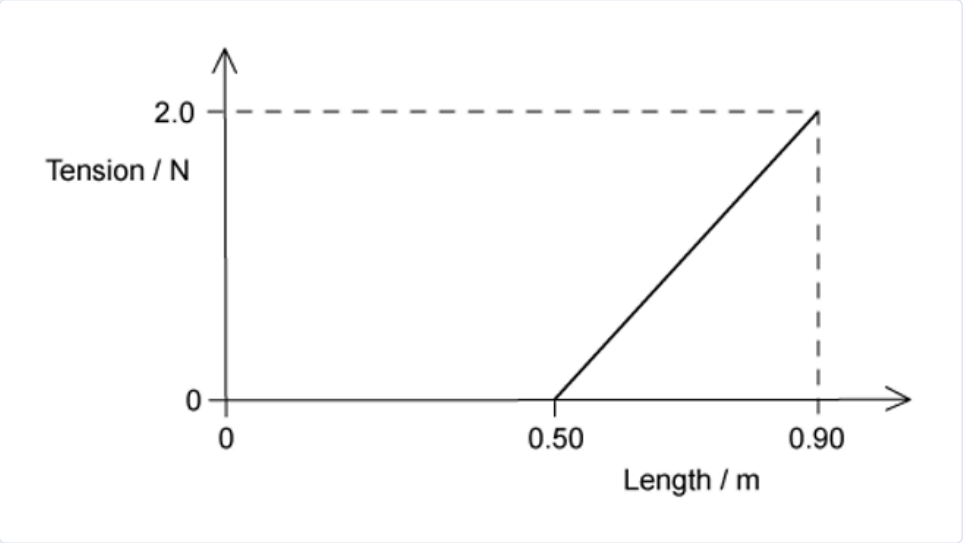
**C** 85 mJ

**D** 5.8 mJ

**Question 19** Medium [1 mark]

A force of 2.0 N is applied to a spring of unextended length 0.50 m and has a new length of 0.90 m.

A graph shows the variation of its length with tension.




How much strain energy is stored in the spring ?

**A** 1.8 J

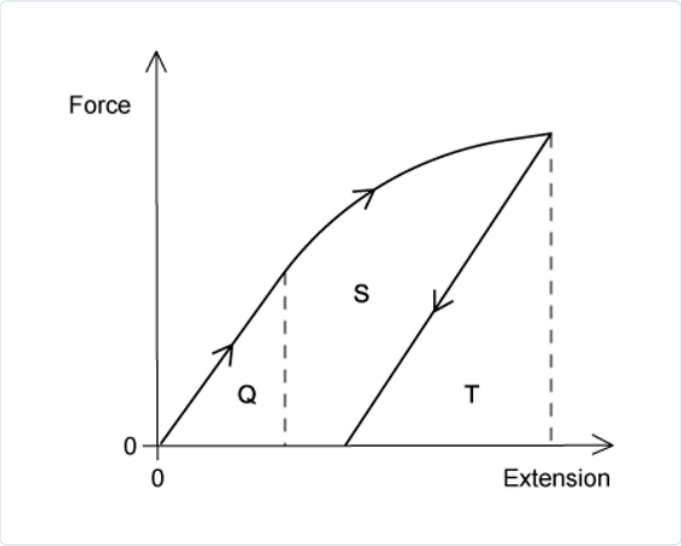
**B** 0.90 J

**C** 0.40 J

**D** 0.80 J

**Question 20** Medium [1 mark]

A student carried out an experiment on a ductile material; stretching it with a tensile force beyond its elastic limit. The force is removed.




The graph shows the force-extension of this experiment.

Which area represents the net work done on the sample?

**A** *T*

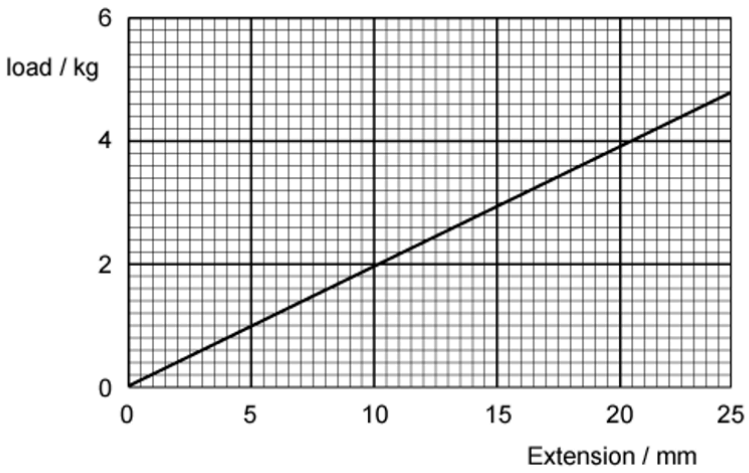
**B** *S* + *T*

**C** *Q* + *S*

**D** *Q*

**Question 21** Hard [1 mark]

A wire undergoes elastic deformation; the graph below shows the load-extension graph.




How much work is done on the wire to increase the extension from 10 mm to 20 mm?

**A** 0.18 J

**B** 0.28 J

**C** 0.37 J

**D** 0.59 J

**Question 22** Hard [1 mark]

A material with elastic properties that obeys Hooke's law has a Young modulus *E* and has a tensile stress of *S* applied.

What is the expression for the elastic energy stored per unit volume of the material?

**A** $\frac{E}{2s^2}$

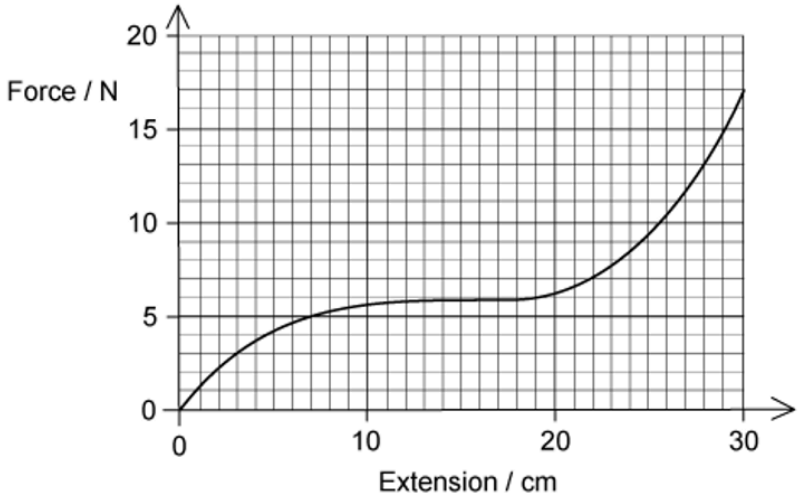
**B** $\frac{S^2}{E}$

**C** $\frac{S^2}{2E}$

**D** $\frac{2E}{S^2}$

**Question 23** Hard [1 mark]

A graph was drawn of the effect on the length of applying a force to a rubber cord.




What is the maximum strain energy in this deformed rubber cord?

**A** 200 J

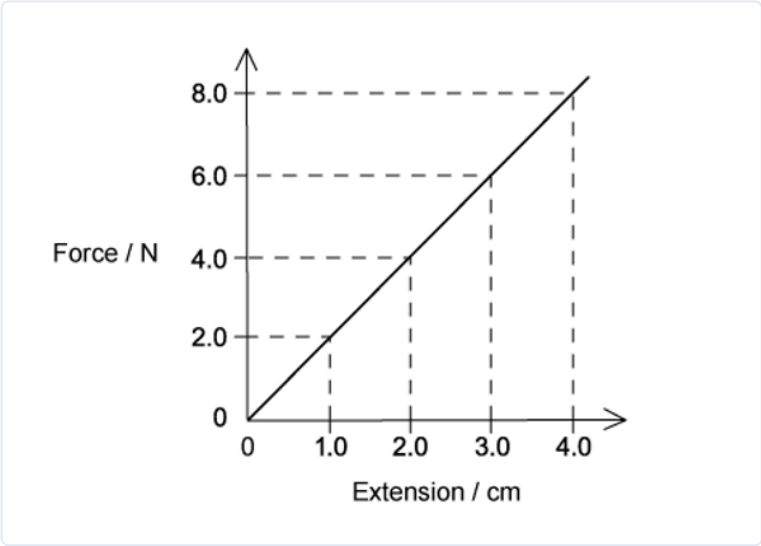
**B** 5.1 J

**C** 2.5 J

**D** 1.9 J

**Question 24** Hard [1 mark]

The graph below shows how force changes the extension of a spring.




The length of the spring with no force applied was 1 cm.

What is the **increase** in the strain energy stored in the spring when its **length** is increased from 2.0 cm to 3.0 cm ?

**A** 0.050 J

**B** 0.040 J

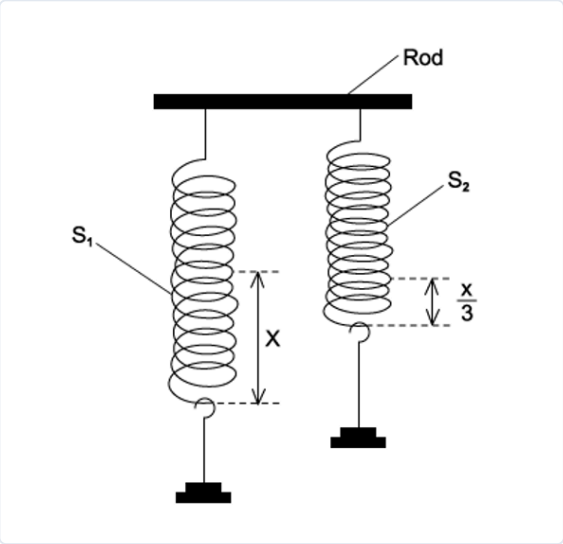
**C** 0.030 J

**D** 0.020 J

**Question 25** Hard [1 mark]

Two springs are attached to a rod, $S_1$ and $S_2$. A load is applied to the bottom of $S_1$, and it extends by $x$ with an elastic potential energy in the spring of $E$.

The same load is applied to the bottom of $S_2$. The extension of $S_2$ is $\frac{x}{3}$, as shown in the diagram




Which equation shows the elastic potential energy in $S_2$?

**A** $9E$

**B** $3E$

**C** $\frac{E}{3}$

**D** $\frac{E}{9}$